SURICATA IDS
PRACTICAL LAB MANUAL

Expanded portfolio edition • Screenshot evidence • Troubleshooting • GitHub workflow

Authorized lab use only. Do not test public or third-party systems.

Purpose

Use Suricata to detect a TCP SYN connection attempt to port 8080 in an authorized isolated lab. This lab covers setup, a custom rule, Netcat traffic, log analysis, screenshot evidence, troubleshooting, and reporting.

Learning objectives

• Identify the correct network interface and addresses.

• Install and validate Suricata.

• Load and test a local signature.

• Generate TCP traffic with Netcat.

• Analyze fast.log and eve.json.

• Capture clear, authentic screenshots for a portfolio.

Topology

Client VM (<CLIENT_IP>) -> isolated lab network -> Suricata sensor (<SURICATA_IP>), monitored on <INTERFACE>.

Prerequisites

• Ubuntu/Debian-like Linux VM(s), sudo on sensor.

• Client and sensor connected to the same intended lab network.

• Suricata, jq, tcpdump on sensor; Netcat on client.

Step 1 — Identify interface

```
ip -br address
ip route
sudo suricata --list-interfaces
```

Use the interface through which the test packets arrive. ip route get <CLIENT_IP> can help determine the route.

Step 2 — Install

```
sudo apt update
sudo apt install -y suricata jq tcpdump
suricata --build-info
```

Client:

```
sudo apt update
sudo apt install -y netcat-openbsd
```

Step 3 — Add and validate rule

```
sudo install -d /etc/suricata/rules
sudo cp suricata/rules/local.rules /etc/suricata/rules/local.rules
sudo suricata -T -c /etc/suricata/suricata.yaml
```

Ensure the active suricata.yaml includes local.rules in rule-files. Merge settings into the existing file; do not overwrite the whole configuration.

Rule:

```
alert tcp any any -&gt; any 8080 (msg:"LAB TCP 8080 connection attempt"; flags:S; sid:1000001; rev:1;)
```

This alerts on TCP SYN packets to destination port 8080. It is a simple educational signature.

Step 4 — Start Suricata

```
sudo systemctl status suricata --no-pager
sudo systemctl restart suricata
```

If manually choosing an interface, stop the service first:

```
sudo systemctl stop suricata
sudo suricata -c /etc/suricata/suricata.yaml -i &lt;INTERFACE&gt;
```

Do not run competing instances against the same log files.

Step 5 — Listener and client traffic

Sensor:

```
nc -lv &lt;SURICATA_IP&gt; 8080
```

Client:

```
nc -v &lt;SURICATA_IP&gt; 8080
```

Type LAB-TEST-8080 and press Enter. Start the listener before connecting. Netcat options differ between versions; check nc -h if needed.

Step 6 — Analyze logs

```
sudo grep 'LAB TCP 8080' /var/log/suricata/fast.log
sudo tail -n 20 /var/log/suricata/eve.json | jq .
sudo jq 'select(.event_type == "alert" and .alert.signature_id == 1000001)' /var/log/suricata/eve.json
```

Expected: a custom alert with SID 1000001. Actual IPs and ephemeral ports vary. If no alert appears, confirm the rule is loaded, the correct interface is monitored, and traffic reaches it.

Step 7 — Screenshot evidence

Follow SCREENSHOT_GUIDE.md. Capture actual screenshots of topology, addresses, rule validation, rule text, running sensor, Netcat connection, fast.log alert, EVE JSON, and final checklist. Store them in evidence/. The sample logs are synthetic examples and must not be presented as real capture evidence.

Step 8 — Troubleshooting

• No route to host: check VM network mode, addresses, subnet, routes, and firewall.

• Connection refused: start listener and check ss -lntp | grep 8080.

• Wrong interface: check ip route get <CLIENT_IP> and tcpdump -ni <INTERFACE> tcp port 8080.

• Missing alerts: run suricata -T, ensure local.rules is loaded, start Suricata first, then reconnect.

• Missing JSON alerts: ensure EVE alert output is enabled in the active configuration.

Step 9 — Checklist

■ IPs and interface documented.

■ Configuration test succeeds.

■ Rule loaded.

■ Listener active.

■ Traffic visible at sensor.

■ SID 1000001 appears in fast.log or eve.json.

■ Screenshots captured and saved.

■ Results template completed.

■ Sensitive data redacted.

Student questions

• What is the difference between IDS and IPS?

• Why does interface selection matter?

• What does flags:S match?

• What are sid and rev?

• Why might tcpdump see traffic while Suricata does not alert?

• Which EVE fields identify the flow?

• How could you constrain the rule to a specific source?

• Why is connection success different from detection success?

Practical challenge

Create a TCP SYN rule for port 9090 with SID 1000002, validate it, generate traffic only in the authorized lab, capture the alert, and record results in results/RESULTS_TEMPLATE.md.